Kristof Möller

Biochemist

Date of birth:	**13.03.1995**
nationality:	**german**
Institutional E-Mail:	**kr.moeller@iaea.org**
Phone number:	**(+377) 97 97 72 59**
ORCID:	**0000-0002-4590-6223**

Current professional activity:

Associate Research Scientist investigating Harmful Algal Blooms and Biotoxins and managing technical cooperation projects in the Radioecology Laboratory of the IAEA Marine Environment Laboratories – Department of Nuclear Sciences and Applications since the 01.11.2024.

Degrees

2024 – University Bremen, Germany / private scholarship from the Deutsche Bundesstiftung Umwelt

Dr. rer. nat. – Doctor of natural sciences – Final grade: *magna cum laude*

2020 – Ludwig-Maximilians-University Munich, Germany

M Sc. - Master of Science in Biochemistry (English degree) – Final grade: 1.5

Major fields: Biochemistry and organic chemistry

2017 – Ludwig-Maximilians-University Munich, Germany

B. Sc. - Bachelor of Science in Chemistry and Biochemistry – Final grade: 2.1

Major fields: Inorganic, organic and physical chemistry

2013 – Albert-Einstein-Schule, Schwalbach am Taunus, Germany

Major fields: Mathematics and Chemistry – Final grade: 1.5

Included a scholar year abroad in New Zealand 2010-2011.

Publications

First author:

Möller et al. „Time series analysis of the expansion of *Alexandrium pseudogonyaulax* across Northern European waters." *Submitted to Harmful Algae (2025)*.

Möller et al. "Toxic effects of the emerging *Alexandrium pseudogonyaulax* (Dinophyceae) on four trophic levels of the pelagic food web" Harmful Algae (2024).

Möller et al. "Effects of bottom-up factors on growth and toxin content of the harmful dinoflagellate *Alexandrium pseudogonyaulax*." Limnology & Oceanography (2024).

Möller et al. "Method optimization of the simultaneous detection of B_{12} congeners leading to the detection of a novel isomer of hydroxycobalamin in seawater." Rapid Communications in Mass Spectrometry (2022).

Möller et al. “Distribution of phycotoxins in Última Esperanza Province during the PROFAN expedition 2019.” Progress in Oceanography (2022).

Co-Author:
Haut et al. “A Negishi cross-coupling reaction enables the total synthesis of (+)-stachyflin.” Tetrahedron (2018).

Sciences

2021 – 2024 – Alfred-Wegener-Institute for Polar and Marine Sciences, Bremerhaven

Phd-student – Expansion factors of the toxic and potentially fish toxic microalgae *Alexandrium pseudogonyaulax* in German coastal waters and assessment of the future hazard potential

Sterile cultivation of various organisms (microalgae, zooplankton); Risk assessment of marine toxins through toxicological cell assays; Extensive time-series analysis of long-term monitoring data from several Northern European countries in international collaboration (Denmark, Sweden, Norway); Publication of scientific results in both journals and conferences; Logistic preparation and execution of ship expeditions. Supervised by PD Dr. Cédric Meunier and co-supervised by Dr. Urban Tillmann, Dr. Bernd Krock and Dr. Florian Koch.

2020 – Alfred-Wegener-Institute for Polar and Marine Sciences, Bremerhaven

Master thesis - Optimization of Vitamin B_{12} Analysis in Seawater using LC-MS/MS

Successful establishment of a vitamin B_{12} extraction method from seawater; Optimization and enhancement of sensitivity throughout the entire analytical process including liquid chromatography and triple quadrupole mass spectrometry. Supervised by Prof. Dr. Klaus Förstemann and co-supervised by Dr. Bernd Krock and Dr. Florian Koch.

2017 – Ludwig-Maximilians-University, Munich

Bachelor thesis - Towards the Bioinspired total Synthesis of Stachyflin and Aureol

Independent contribution to an extensive organic synthesis of two natural compounds and autonomous purity determination via NMR and mass spectrometry; complex handling of toxic and hydrolysis-sensitive substances under inert gas. Supervised by Prof. Dr. Thomas Magauer.

Fieldwork and Research Stays

2023 - Aurora –Baltic Sea – AWI in cooperation with Aarhus University, Denmark (1 week)

Logistical preparation and successful execution of laboratory work in a multinational team.

2023 - Alkor – Baltic Sea – University Hamburg (1 week)

Logistical preparation and successful execution of laboratory work; introduction of master students from the University of Hamburg to the sampling of toxic microalgae.

2023 - Wien – Universität Wien (4 weeks)

Toxicological risk assessment of marine toxins; sterile work with sensitive cell lines; results yielded the best presentation award at the ICHA conference.

2022/2023 - Roskilde – Aarhus University, Roskilde, Denmark (3 weeks)

Diverse time series analysis in R: Independent formatting, filtering, and visual representation of ecological datasets; publication of the results in my dissertation and presentation at a scientific conference.

2022 - Black Sea – AWI in cooperation with TÜBITAK Institute (1 week)

Preparation and elaborate execution of laboratory work in a partly non-English-speaking international team on a state-owned Turkish research vessel.

2020 - Uthörn – North Sea, Baltic Sea (4 weeks)

Successful completion of various laboratory tasks and work within a multinational team. Responsible for overall scientific operations aboard the ship as expedition leader, coordinating schedules in collaboration with the ship's captain (total duration: 10 days).

Conferences

2024 - Ocean Sciences Meeting OSM, USA, New Orleans (Poster)

2023 - International Conference on Harmful Algae ICHA, Japan, Hiroshima (Presentation)

ISSHA Maureen Keller Award for the best oral student presentation

2021 - ICYMARE, Germany, Bremerhaven (Presentation)

Other professional experiences

2014 - 2020 – Staff at Jochen Schweizer Erlebnisbetriebe (HEAG GmbH)
Co-Teamcaptain (2015-2018) and Teamcaptain (2018-2020) at Bungee-Jumping stations:
Main responsible for the safe and organized execution of events.

Andere Aktivitäten

2021 - 2022: Boulder/Climbing-Trainer for Kids in the Bremerhavener Boulder Club

2021 - Member of the doctoral student representation, Alfred-Wegener-Institute, Germany

Skills

Lab

Chemical Analytics (Very good: LC-MS/MS; Good: NMR, IR, UV-Vis, and Raman spectroscopy)

Sterile cell culture work and conducting toxicological assays

Establishment and maintenance of algae and zooplankton cultures

Organic Chemistry and Biochemistry - Advanced research internships in:

- Organic Chemistry (Handling toxic, flammable, and explosive substances; planning and execution of organic syntheses)
- Inorganic Chemistry (Handling toxic substances; crystal structure analyses; various analytical techniques)
- Physical Chemistry (Various analytical techniques and introduction to density functional theory)
- Prebiotic Chemistry (Water chemistry)
- Molecular Genetics (Sterile work with RNA and DNA, cloning, CRISPR-Cas)
- Ribosome Biogenesis (Yeast transformations)

IT-Skills

Very good knowledge: R (statistics and programming language)

Good knowledge: Excel, PowerPoint, LaTeX (word processing program)

Advanced basic knowledge: Python (programming language), Matlab (programming language)

Languages

German: Native language

English: Fluent

French: Very good (B2-C1)

Hobbies

Outdoor activities (climbing, hiking, running), reading